

Peter W. Deutsch

Cambridge – Massachusetts

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Education

Massachusetts Institute of Technology

PhD Candidate, Electrical Engineering and Computer Science

2022 – Present

Doctoral Supervisor: Prof. Mengjia Yan

Massachusetts Institute of Technology

Master of Science, Electrical Engineering and Computer Science

2020 – 2022

Thesis: Mitigating Memory Controller Side-Channels

Masters Supervisor: Prof. Mengjia Yan

University of British Columbia

Bachelor of Applied Science, Computer Engineering

2014 – 2020

Undergraduate Supervisors: Prof. Mieszko Lis & Prof. Prashant Nair

Publications

Vincent Quentin Ulitzsch, Alessandro Bertani, **Peter W. Deutsch**, David Langus Rodriguez, Kelly Xu, Aarti Gupta, Sharad Malik, and Mengjia Yan. INSIGHT: Automatic generation of explanations for efficient identification of hardware bugs and underspecifications. In *Proceedings of the 47th IEEE Symposium on Security and Privacy (S&P)*, 2026.

Peter W. Deutsch, Harish D. Dixit, Gautham Vunnam, Carl Moran, Eleanor Ozer, and Sriram Sankar. PinDrop: Breaking the silence on SDCs in a large-scale fleet. In *Proceedings of the 32nd IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2026.

Peter W. Deutsch*, Vincent Quentin Ulitzsch*, Sudhanva Gurumurthi, Vilas Sridharan, Joel Emer, and Mengjia Yan. DelayAVF: Calculating architectural vulnerability factors for delay faults. In *Proceedings of the 57th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2024.

Peter W. Deutsch, Weon Taek Na, Thomas Bourgeat, Joel S Emer, and Mengjia Yan. Metior: A comprehensive model to evaluate obfuscating side-channel defense schemes. In *Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA)*, pages 1–16, 2023.

Peter W. Deutsch*, Yuheng Yang*, Thomas Bourgeat, Jules Drean, Joel S Emer, and Mengjia Yan. DAGguise: Mitigating memory controller side-channels. In *Proceedings of the 27th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pages 329–343, 2022.

Oliver Willers, Christopher Huth, Jorge Guajardo, Helmut Seidel, and **Peter Deutsch**. On the feasibility of deriving cryptographic keys from MEMS sensors. *Journal of Cryptographic Engineering*, 10(1):67–83, 2020.

Academic Service

IEEE/ACM International Symposium on Microarchitecture (MICRO) <i>Program Committee Member</i>	2026
ACM Transactions on Architecture and Code Optimization (TACO) <i>Reviewer</i>	2026
IEEE Transactions on Computers <i>Reviewer</i>	2026
IEEE International Symposium on High-Performance Computer Architecture (HPCA) <i>External Review Committee Member</i>	2025
IEEE Transactions on Computers – Special Issue on Hardware Security <i>Reviewer</i>	2022

Invited Talks

Machine Learning Accelerator Reliability: Assessing Faults During Training <i>Advanced Micro Devices (AMD) - Architecting for Marginal Defects Working Group</i>	May 2025
DelayAVF: Calculating Architectural Vulnerability Factors for Delay Faults <i>Advanced Micro Devices (AMD) - Architecting for Marginal Defects Working Group</i>	May 2024
Increasing Architectural Resilience to Small Delay Faults <i>MIT AI Hardware Program Symposium</i>	May 2024
Strengthening Hardware Security in the Age of AI <i>MIT CSAIL/UK GCHQ Delegation</i>	Sept. 2023
Modelling Obfuscating Side-Channel Defense Schemes <i>Carnegie Mellon University - Special Topics in Hardware Security (17-715)</i>	Sept. 2023

Work Experience

Research & Academic.....

Meta <i>PhD Intern</i>	Sunnyvale, CA 2025
• Analyzed long-term silent data corruption (SDC) behaviours across Meta's CPU fleet. • Developed reliability forecasting models for hyper-scale computing platforms, helping to scale up next-generation LLM training.	
Advanced Micro Devices (AMD) <i>PhD Intern</i>	Austin, TX 2024
• Applied my prior research (DelayAVF) to assess the potential reliability impacts of small delay faults in next-generation AMD products. • Substantially expanded upon the existing open-source DelayAVF infrastructure to facilitate the reliability assessment of large out-of-order CPUs.	

Massachusetts Institute of Technology**Cambridge, MA***Teaching Assistant/Lab Assignment Developer**2022 – 2024*

- Assisted in the development and deployment of lab assignments for MIT's Secure Hardware Design course.
- Developed an assignment which guides students through performing and characterizing Rowhammer attacks on commodity hardware.

University of British Columbia**Vancouver, Canada***Undergraduate Research Student**2019 – 2020*

- Investigated methods to detect and mitigate speculative execution attacks which utilize cache and DRAM side-channels (ex. Spectre/Meltdown).
- Replicated attacks, benchmarked prior work, and explored new mitigations using SPEC CPU 2017 and gem5.

Bosch Corporate Research**Stuttgart, Germany***Microsystems Engineering Student**2017*

- Researched the use of MEMS gyroscopes as Physical Unclonable Functions (PUFs), facilitating reliable secret key generation in IoT devices.
- Helped to devise and evaluate entropy extraction schemes to generate cryptographically secure keys from highly correlated device features.

University of British Columbia**Vancouver, Canada***Undergraduate Teaching Assistant**2016 – 2020*

- Conveyed Verilog-focused digital design content to hundreds of second and third-year undergraduate students.
- Taught CPEN 211 (Introduction to Microcomputers), CPEN 311 (Digital Systems Design), and CPEN 391 (Computer Engineering Design Studio II).

Industry**Intel Corporation****Vancouver, Canada***Verification Engineer Intern**2018 – 2019*

- Verified system controller ASICs for Intel NAND devices using SystemVerilog and the Universal Verification Methodology (UVM).
- Designed end-to-end traffic tests to confirm compliance with internal architecture requirements and flash interface specifications, ensuring that comprehensive code coverage was achieved.

Microsemi (Microchip)**Vancouver, Canada***Product Design Engineer Intern**2017*

- Designed and verified top-level RTL glue logic (SystemVerilog & VHDL) for SAS/SATA RAID controllers.
- Implemented appropriate pipelining and clock-domain-crossing synchronization strategies, ensuring that timing closure and MTBF thresholds were met.

Awards**Graduate****HPCA Best Paper Candidate (PinDrop)****2026***Awarded to the best 4 (out of 119) papers accepted to HPCA in 2026***IEEE Micro Top Picks Honourable Mention (DelayAVF)****2025***Awarded to papers deemed the most significant in computer architecture research in 2024***Google Research Scholar Grant****2023***Topic: Leveraging Accessible Signals for the Efficient Discovery of Corrupt Execution Errors***Advanced Television and Signal Processing Fellowship****2020***Awarded on the recommendation of the Department Head of EECS*

Undergraduate.....

Dean's Prize for Academic Excellence in Engineering	2020
<i>Awarded to the head of the graduating undergraduate class in Applied Science</i>	
ECE Capstone Faculty Award	2020
<i>Presented to the top ECE Capstone (final year) project teams in 2020</i>	
NSERC Undergraduate Student Research Award	2019
<i>Awarded on the recommendation of the Faculty of Applied Science</i>	
Trek Excellence Scholarship for Continuing Students	2015, 2016, 2017, 2019
<i>Awarded to students in the top 5% of their program</i>	
PMC-Sierra Founders Award in Electrical and Computer Engineering	2019
<i>Awarded on the recommendation of the Department Head of Computer Engineering</i>	
Elizabeth and Leslie Gould Scholarship in Engineering	2019
<i>Awarded on the recommendation of the Faculty of Applied Science</i>	
J Fred Muir Memorial Scholarship in Engineering	2017
<i>Awarded on the recommendation of the Faculty of Applied Science</i>	
J K Zee Memorial Scholarship	2016
<i>Awarded on the recommendation of the Faculty of Applied Science</i>	

Volunteerism

MIT Graduate Application Assistance Program	Cambridge, MA
<i>Treasurer/Graduate Student Volunteer</i>	<i>2021 – Present</i>
• Worked with underrepresented MIT PhD applicants, providing advice and detailed feedback on personal and research statements. • Coordinated finances for the program, raising funds to provide fee waivers for underprivileged applicants.	
BC COVID-19 3D Printing Group (BCC3D)	Vancouver, Canada
<i>Printing / Distribution Volunteer</i>	<i>2020</i>
• Personally manufactured 300+ 3D printed face shield visors and 'ear savers' for use at hospitals and clinics. • Inspected, sanitized, and packed 10,000+ articles of PPE produced by local volunteers.	